DEBRECENI EGYETEM

project 2

slope stability for a sliding embankment

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subject : geotechnics II

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Part 2 of the design project

Determine the factor of safety against sliding of the embankment with the slices method (simplified Bishop), and for the cut with the block method. With the method of slices, consider 3 different slip circles (base failure, slope failure with positive tangent and slope failure with negative tangent at the slope toe).
According to the design approach DA-3 of Eurocode 7, use the design values of the soil parameters to determine the safety factor.

Top width of the road:	12 m
Inclination of the pavement:	3,0 %
Height of the embankment (in axis):	9,2 m
Slope of the embankment:	5/4
Height of the cut (in axis):	12,7 m
Slope of cut:	6/4

Characteristic values of soil parameters for the embankment:

Unit weight:	18,0 kN/m^3
Internal friction angle:	28˚
Cohesion:	15 kPa

Characteristic values of soil parameters in cut:

Layer 1 unit weight:	20 kN/m^3
Layer 1 internal friction angle:	26˚
Layer 1 cohesion:	5 kPa
Layer 1 height:	H/4
Layer 2 unit weight:	19,0 kN/m^3
Layer 2 internal friction angle:	30˚
Layer 2 cohesion:	15 kPa
Layer 2 height:	H/2
Inclination of the slip plane:	9˚

Characteristic values of shear strength on the slip surface:

Internal friction angle in the slip surface:	10˚
Cohesion in the slip surface:	10 kPa

Teacher notifications:

	Date	*Signature*
Consultation		
Submission		

second project which is calculation for slope stability

1-Determine the factor of safety against sliding of the embankment with the slices method "simplified Bishop". you should consider 3 different slip circles
• base failure,
• slope failure with positive tangent
• slope failure with negative tangent at the slope toe

• Determine the factor of safety against sliding for the cut with the block method.

for the drawing part:
you should draw a horizontal surface then measure your height then make the road inclination

.draw the slops then find the ground surface intersections after that draw the trences

2-The sliding soil mass is divided vertically in cells (slices)
: 8-10 slices depending on geometry, maximum 1,5 m wide slices

mine was less than 1.5 m width and there was 10 slices for more precision

3-Curved sliding surface can be considered a straight line at the bottom of the slice

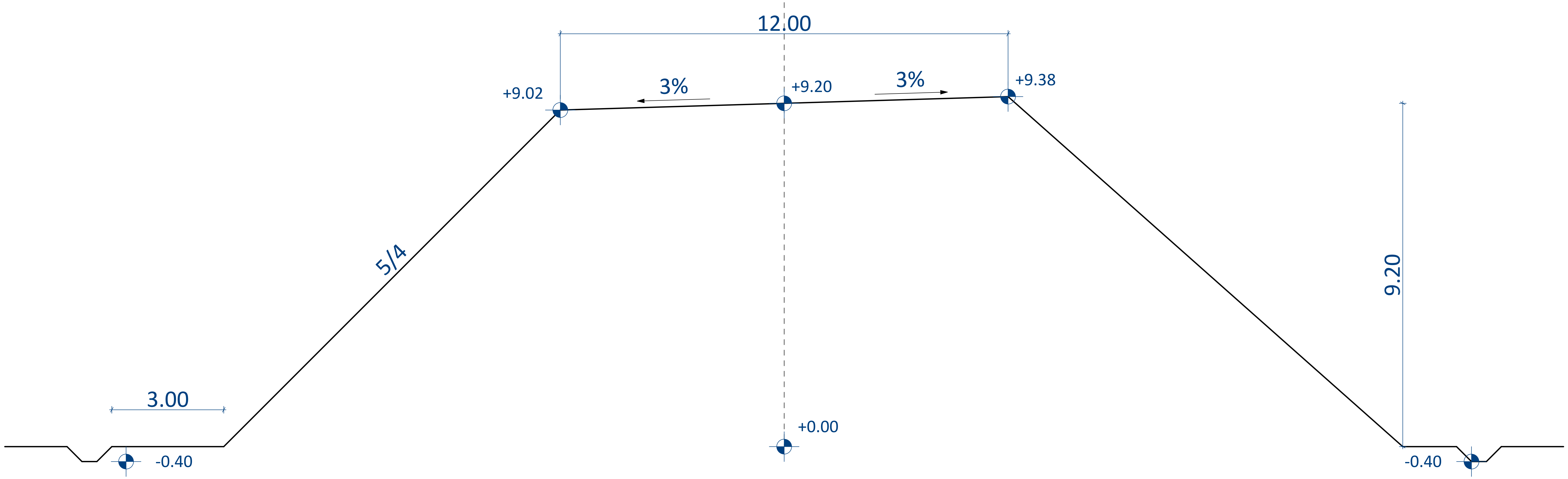
� The inclination α,n is measured

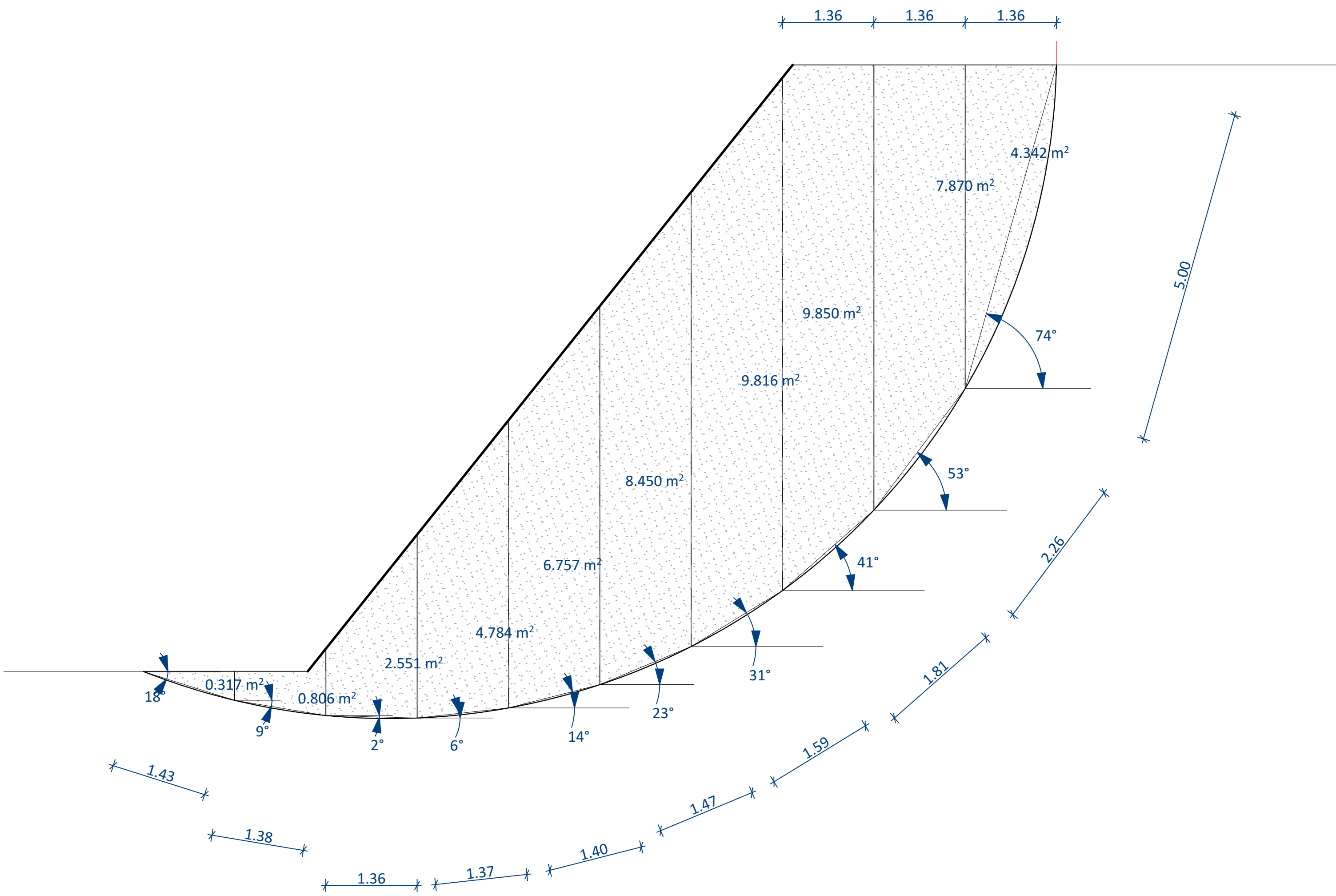
4-For the analysis of the overall stability of a slope and any other geotechnical structure, the design approach DA-3 shall be used, along with the sets A2 „+" M2 „+" R1 of the partial factors.

Soil parameters	Symbol	Set	
		M1	M2 – Slopes [b]
Angle of shearing resistance [a]	$\gamma_{\varphi'}$	1,0	1,35
Effective cohesion	$\gamma_{c'}$	1,0	1,35
Undrained shear strength	γ_{cu}	1,0	1,5
Unconfined strength	γ_{qu}	1,0	1,5
Unit weight	γ_{γ}	1,0	1,0

5-Tangent at the intersection with the crest should not curved backwards (maximum vertical)!
✓ The intersection at the toe (case a) also your decision!

One-sided slope





A								
Slice	*Area (m2)*	*Self weight of the slice (kN/m3)*	*Apha angle (°)*	*Force parallel to the sliding surface (Tn)*	*Forces normal to the sliding surface*	*Length (L)*	*Cohesion along the sliding surface*	*Friction from the self weight and friction angle*
	$[m^2]$	*Wi[kN/m]*	*[o]*	*T_i[kN/m]*	*N_i[kN/m]*	*L_i[m]*	*C_i[kN/m]*	*Fi[kN/m*
1	0.317	5.71	-18	-1.76	5.43	1.43	15.89	2.14
2	0.806	14.51	-9	-2.27	14.33	1.38	15.33	5.64
3	2.551	45.92	-2	-1.6	45.89	1.36	15.11	18.07
4	4.781	86.06	6	9	85.59	1.37	15.22	33.71
5	6.757	121.63	14	29.42	118.01	1.4	15.56	46.48
6	8.45	152.1	23	59.43	140.01	1.47	16.33	55.14
7	9.816	176.69	31	91	151.45	1.59	17.67	59.65
8	9.85	177.3	41	116.32	133.81	1.81	20.11	52.7
9	7.87	141.66	53	113.13	85.25	2.26	25.11	33.58
10	4.342	78.16	74	75.13	21.54	5	55.56	8.48
Sum	55.54	999.74		487.8	801.31	19.07	211.89	315.59

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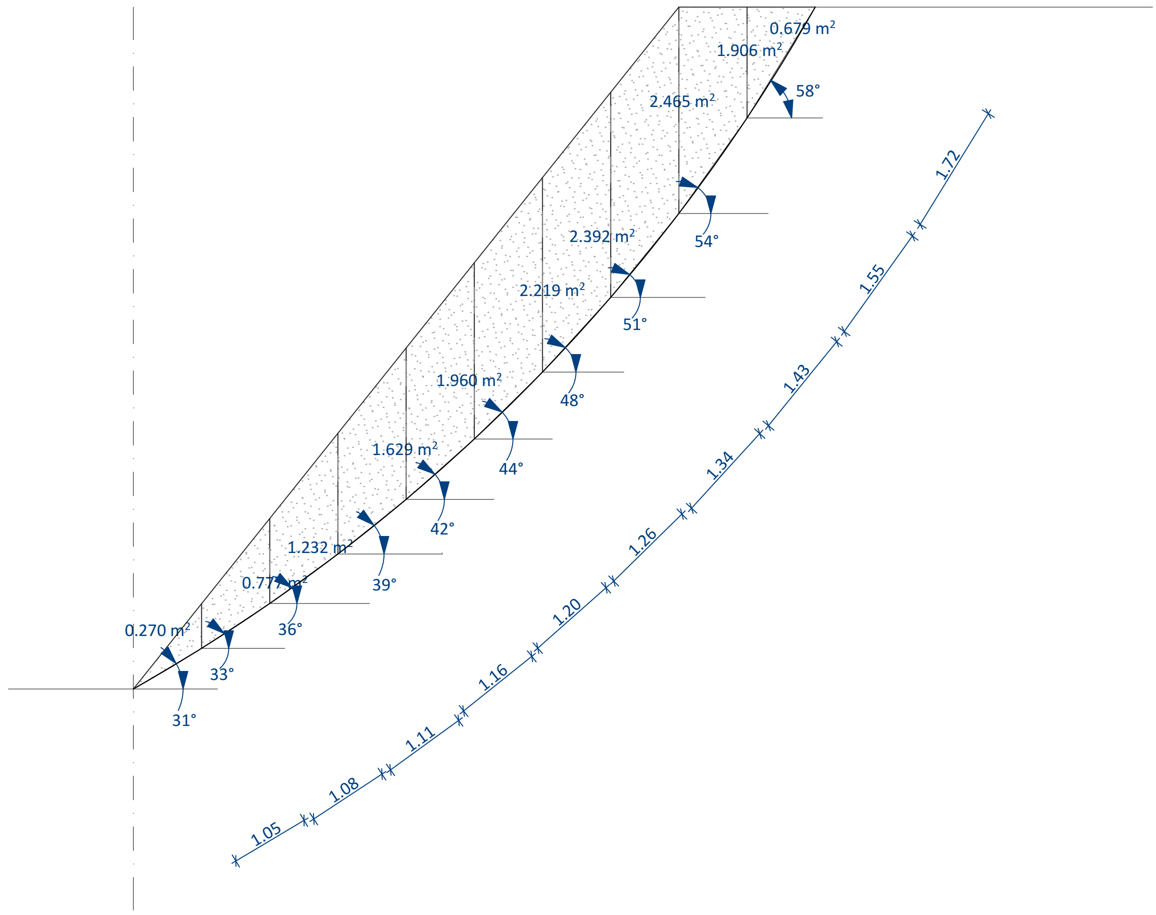

B

Slice	*Area (m2)*	*Self weight of the slice (kN/m3)*	*Apha angle (°)*	*Force parallel to the sliding surface (Tn)*	*Forces normal to the sliding surface*	*Length (L)*	*Cohesion along the sliding surface*	*Friction from the self weight and friction angle*
	[m^2]	*Wi[kN/m]*	*[o]*	*T_i[kN/m]*	*N_i[kN/m]*	*L_i[m)*	*C_i[kN/m]*	*Fi[kN/m*
1	0.27	4.86	31	2.50	4.17	1.05	11.67	1.64
2	0.77	13.86	33	7.55	11.62	1.08	12.00	4.58
3	1.232	22.176	36	13.03	17.94	1.11	12.33	7.07
4	1.629	29.322	39	18.45	22.79	1.16	12.89	8.98
5	1.96	35.28	42	23.61	26.22	1.2	13.33	10.33
6	2.219	39.942	44	27.75	28.73	1.26	14.00	11.32
7	2.392	43.056	48	32.00	28.81	1.34	14.89	11.35
8	2.465	44.37	51	34.48	27.92	1.43	15.89	11.00
9	1.906	34.308	54	27.76	20.17	1.55	17.22	7.94
10	0.679	12.222	58	10.36	6.48	1.72	19.11	2.55
Sum	**15.522**	**279.396**		**197.492**	**194.844**	**12.900**	**143.332**	**76.741**

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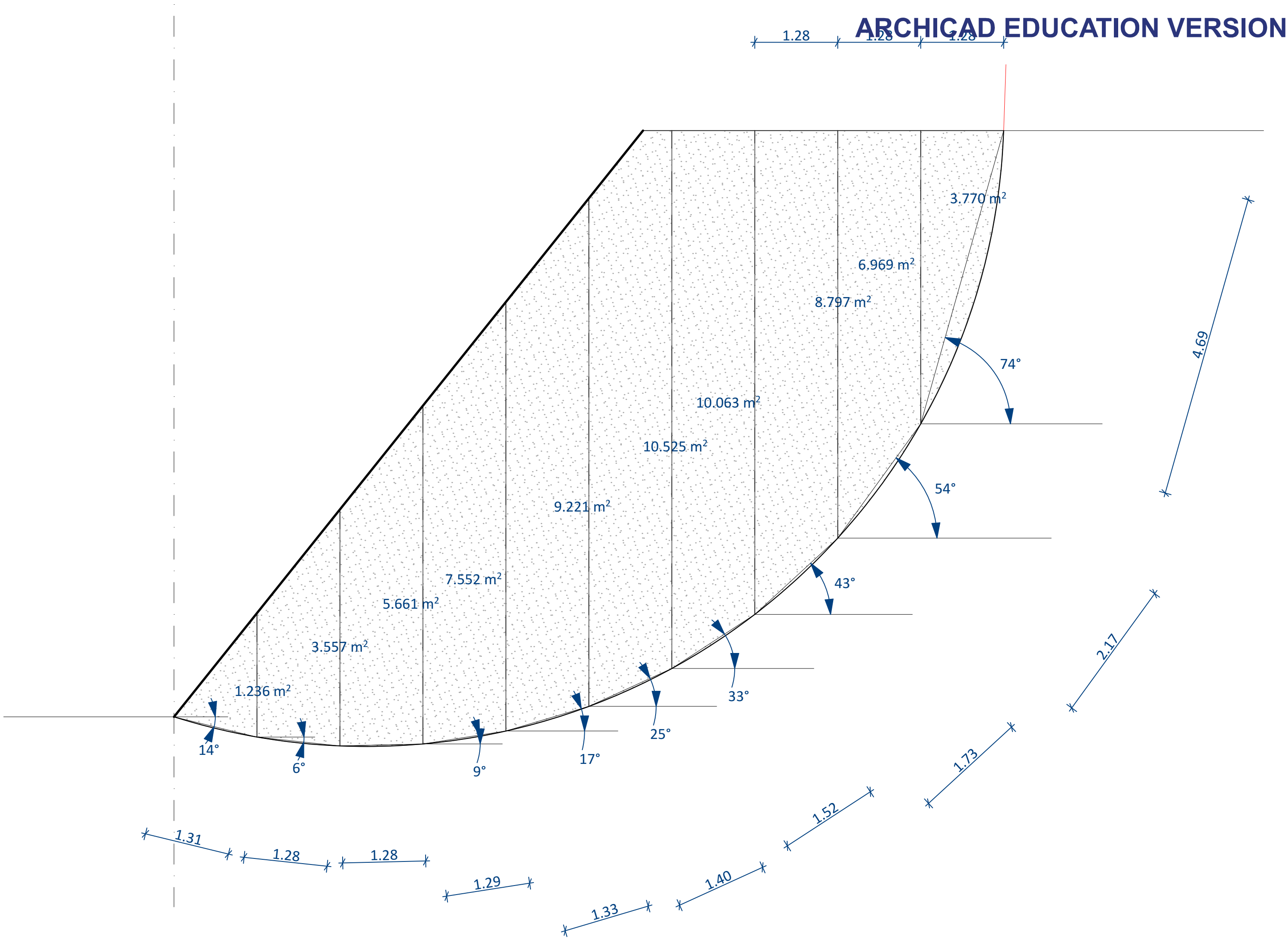

C

Slice	Area (m2)	Self weight of the slice (kN/m3)	Apha angle (˚)	Force parallel to the sliding surface (Tn)	Forces normal to the sliding surface	Length (L)	Cohesion along the sliding surface	Friction from the self weight and friction angle
	[m^2]	Wi[kN/m]	[o]	T_i[kN/m]	N_i[kN/m]	ΔL_i[m]	C_i[kN/m]	Fi[kN/m]
1	1.236	22.25	-14	-5.38	21.59	1.31	14.56	8.5
2	3.557	64.03	-6	-6.69	63.68	1.28	14.22	25.08
3	5.661	101.9	0	0	101.9	1.28	14.22	40.13
4	7.552	135.94	9	21.27	134.26	1.29	14.33	52.88
5	9.221	165.98	17	48.53	158.73	1.33	14.78	62.52
6	10.525	189.45	25	80.07	171.7	1.4	15.56	67.63
7	10.063	181.13	33	98.65	151.91	1.52	16.89	59.83
8	8.797	158.35	43	107.99	115.81	1.73	19.22	45.61
9	6.969	125.44	54	101.48	73.73	2.17	24.11	29.04
10	3.77	67.86	74	65.23	18.7	4.69	52.11	7.37
Sum	67.351	1212.33		511.15	1012.01	18	200	398.59

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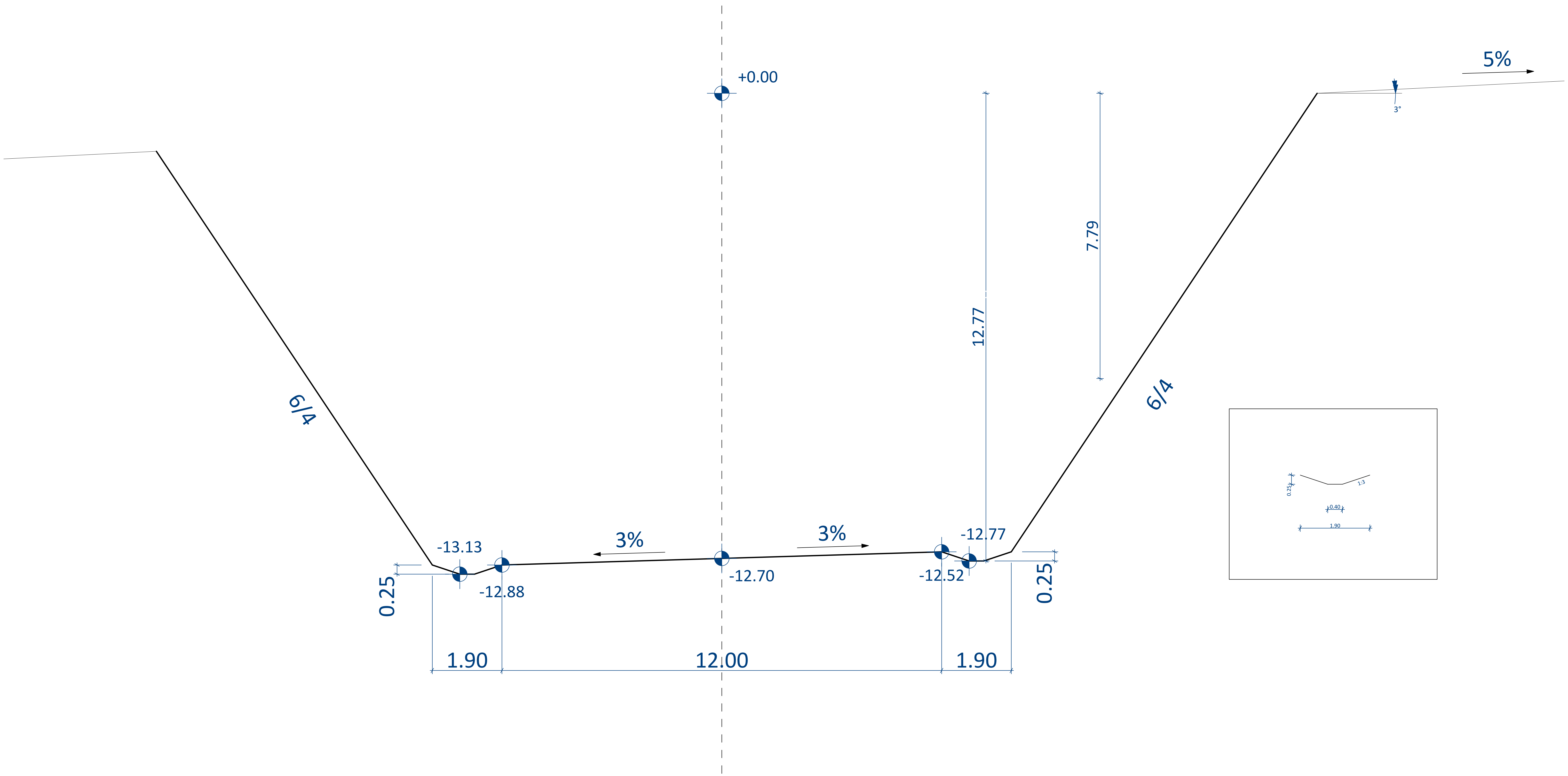
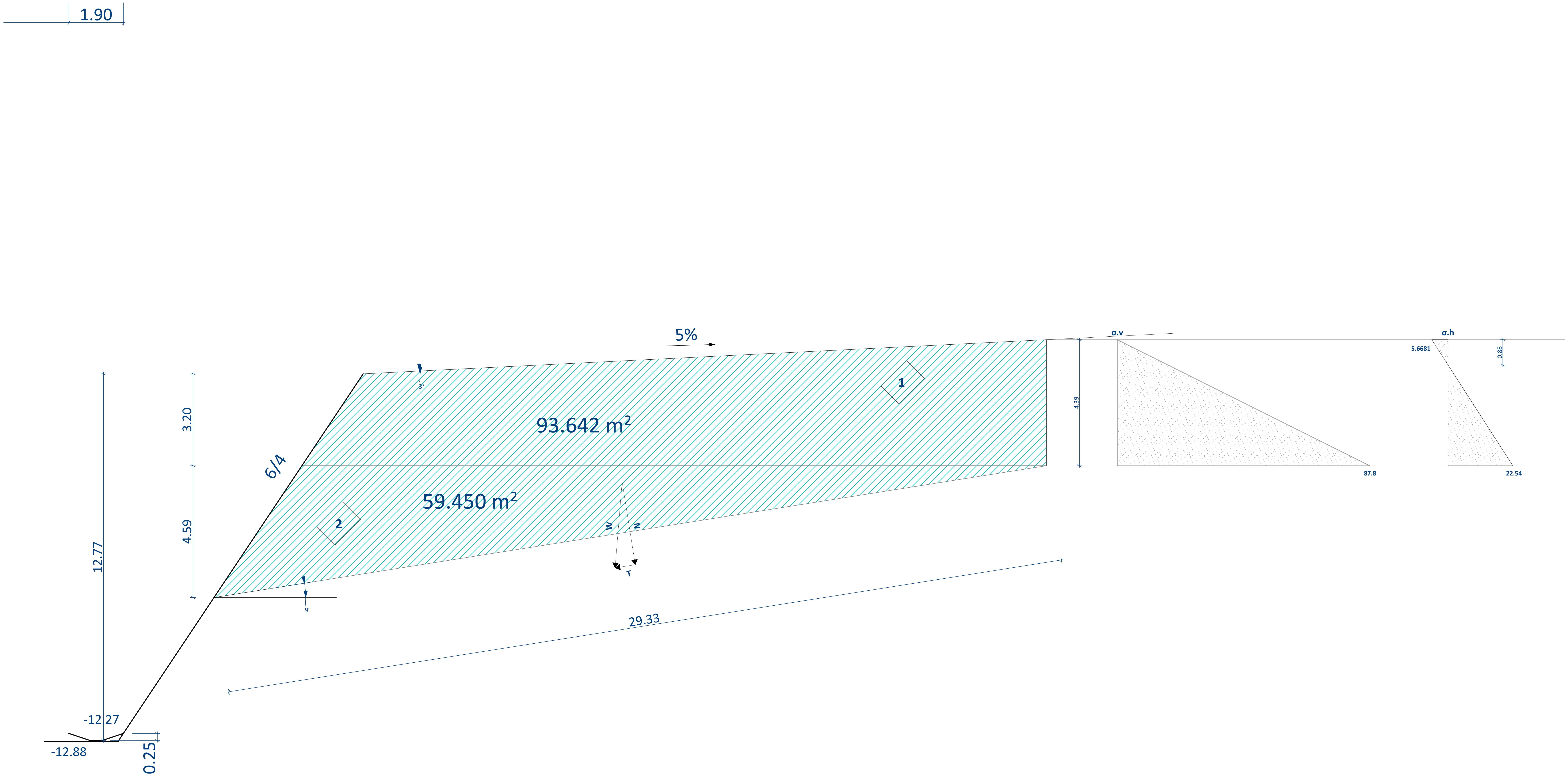
One-sided slope




1.90
5%
σ.v
σ.h
5.6681
0.88
4.39
1
93.642 m2
3.20
6/4
12.77
59.450 m2
2
4.59
W
N
T
87.8
22.54
3°
9°
29.33
-12.27
-12.88
0.25

instruction for drawing the cut and solving the calculations

1-The sliding is on the surface of a weak layer

2-start by drawing the sliding surface from getting the ε and then connect the slip surface and the horizontal surface to create the huge block then add up the areas

3-The angle of the back-face of the wall in the expression is α=0°,
the angle of wall friction” δ=φ

4-β (inclination of ground surface: β=5% to the left)

5-because of cohesion there might be a negative part in the stress diagram so consider only the positive area

the cut calculations

$\varphi_1 := 26\ \mathrm{deg}$ $\gamma_1 := 20\ \frac{\mathrm{kN}}{\mathrm{m}^3}$ $C_1 := 5\ \mathrm{kPa}$ $H := 12.77\ \mathrm{m}$

from B-C (the slip surface)

$\varepsilon := 9\ \mathrm{deg}$ $h_2 := \frac{H}{2} = 6.385\ \mathrm{m}$ $\gamma_2 := 19\ \frac{\mathrm{kN}}{\mathrm{m}^3}$ $\varphi_2 := 30\ \mathrm{deg}$ $c_2 := 15\ \mathrm{kPa}$

$A := (93.642 + 59.45)\ \mathrm{m}^2$ $L := 29.33\ \mathrm{m}$

C.SL.d is the effective cohesion on the sliding surface $C'_{SL.k} := 10\ \mathrm{kPa}$

φ.csl.d is the effective angle of shearing resistance on the sliding surface $\varphi_{csl.k} := 10\ \mathrm{deg}$

constants

$\gamma_{c'} := 1.35$ $\gamma_{\varphi'} := 1.35$

Soil parameters	Symbol	Set M1	Set M2 – Slopes [b]
Angle of shearing resistance [a]	$\gamma_{\varphi'}$	1,0	1,35
Effective cohesion	$\gamma_{c'}$	1,0	1,35
Undrained shear strength	γ_{cu}	1,0	1,5
Unconfined strength	γ_{qu}	1,0	1,5
Unit weight	γ_{γ}	1,0	1,0

1/ Active earth pressure coefficient:

$\delta := \varphi_1$ $\alpha := 0\ \text{deg}$ $\beta := 3\ \text{deg}$

$$K_a := \frac{\cos(\delta)\cdot\left(\cos\left(\varphi_1+\alpha\right)\right)^2}{\left(\cos(\alpha)\right)^2\cdot\left(\sqrt{\cos(\alpha-\delta)}+\sqrt{\frac{\sin\left(\varphi_1+\delta\right)\cdot\sin\left(\varphi_1-\beta\right)}{\cos(\alpha+\beta)}}\right)^2} = 0.3213$$

from A-B

$h_1 := \frac{H}{4} = 3.1925\ \text{m}$ $h := 4.39\ \text{m}$

$\sigma'_v := h\cdot\gamma_1 = 87.8\ \text{kPa}$

2/ The horizontal stress:

at A

$\sigma'_{h.a1} := -2\cdot C_1\cdot\sqrt{K_a} = -5.6681\ \text{kPa}$

when the horizontal stress diagrma will be zero since this is an active case with load on top

$Z_0 := 0.8821\ \text{m}$ you can get this from the graph or this equation

$\frac{x}{5.6681} = \frac{4.39-x}{22.54}$ $x := \frac{4.39\cdot5.6681}{22.54+5.6681} = 0.8821$

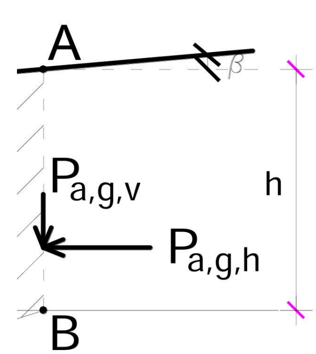
at the end of h

$\sigma'_{h.a} := \sigma'_v\cdot K_a - 2\cdot C_1\cdot\sqrt{K_a} = 22.54\ \text{kPa}$

3/ Then you calculate the Earth pressure force Pa,g,h from the positive area

$$P_{a.g.h} := \frac{1}{2}\cdot\left(K_a\cdot\gamma_1\cdot h - 2\cdot C_1\cdot\sqrt{K_a}\right)\cdot\left(h-Z_0\right) = 39.5341\ \frac{\text{kN}}{\text{m}}$$

4/ The vertical component of the active earth pressure is :



$$P_{a.g.v} := P_{a.g.h}\cdot\tan(\beta) = 2.0719\ \frac{\text{kN}}{\text{m}}$$

5/ The resultant of the active earth pressure force is then :

$$P_{a.g} := \sqrt{\left(P_{a.g.v}\right)^2+\left(P_{a.g.h}\right)^2} = 39.5883\ \frac{\text{kN}}{\text{m}}$$

6/ This have to be divided into components normal (Pa,g,N) and parallel (Pa,g,T) to the slip plane: (ε represents the inclination of the sliding surface)

$$P_{a.g.N} := p_{a.g} \cdot \sin(\beta - \varepsilon) = -4.1381 \ \frac{kN}{m}$$

$$P_{a.g.T} := p_{a.g} \cdot \cos(\beta - \varepsilon) = 39.3715 \ \frac{kN}{m}$$

7/ Self weight of the sliding block :

$$W := A \cdot \gamma_2 = 2908.748 \ \frac{kN}{m}$$

8/ Divided to components normal and parallel to the sliding surface

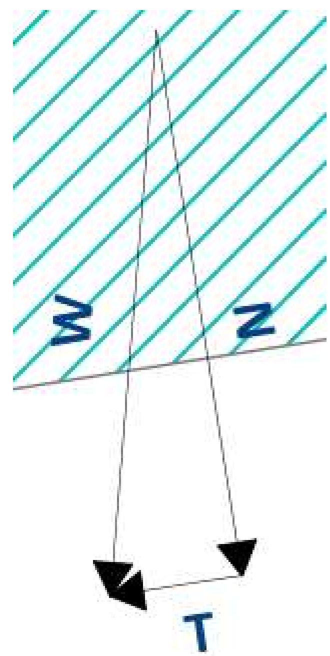
Force parallel to the sliding surface

$$T := W \cdot \sin(\varepsilon) = 455.0284 \ \frac{kN}{m}$$

Forces normal to the sliding surface

$$N := W \cdot \cos(\varepsilon) = 2872.9365 \ \frac{kN}{m}$$

Forces preventing the sliding :



Cohesion along the sliding surface

$$C_{SL.d} := \frac{C'_{SL.k}}{\gamma_{c'}} = 7.4074 \ kPa$$

$$C := C_{SL.d} \cdot L = 217.2593 \ \frac{kN}{m}$$

Friction from the self weight and friction angle

$$\varphi_{csl.d} := \operatorname{atan}\left(\frac{\tan(\varphi_{csl.k})}{\gamma_{\varphi'}}\right) = 7.4414 \ deg$$

$$F := (N + P_{a.g.N}) \cdot \tan(\varphi_{csl.d}) = 374.7012 \ \frac{kN}{m}$$

Forces causing the sliding

$$T + P_{a.g.T} = 494.3999 \ \frac{kN}{m}$$

safety factor is adequate when it is more than 1

$$F_s := \frac{F + C}{T + P_{a.g.T}} = 1.1973$$